```python

def fib_sequence( ceiling):
    """
    compute and return a fibonacci sequence as a list.
    The final element does not exceed the ceiling.
    """
    u = 0
    v = 1
    z_sequence = [ u, v]
    while True:
        w = u + v
        if w > ceiling:
            return z_sequence
        z_sequence.append( w)
        u = v
        v = w
        continue
    return # control never reaches this return

z_sequence = fib_sequence( ceiling = 2000)
print( z_sequence)

"""
The output was:
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55,
 89, 144, 233, 377, 610, 987, 1597]
"""
```